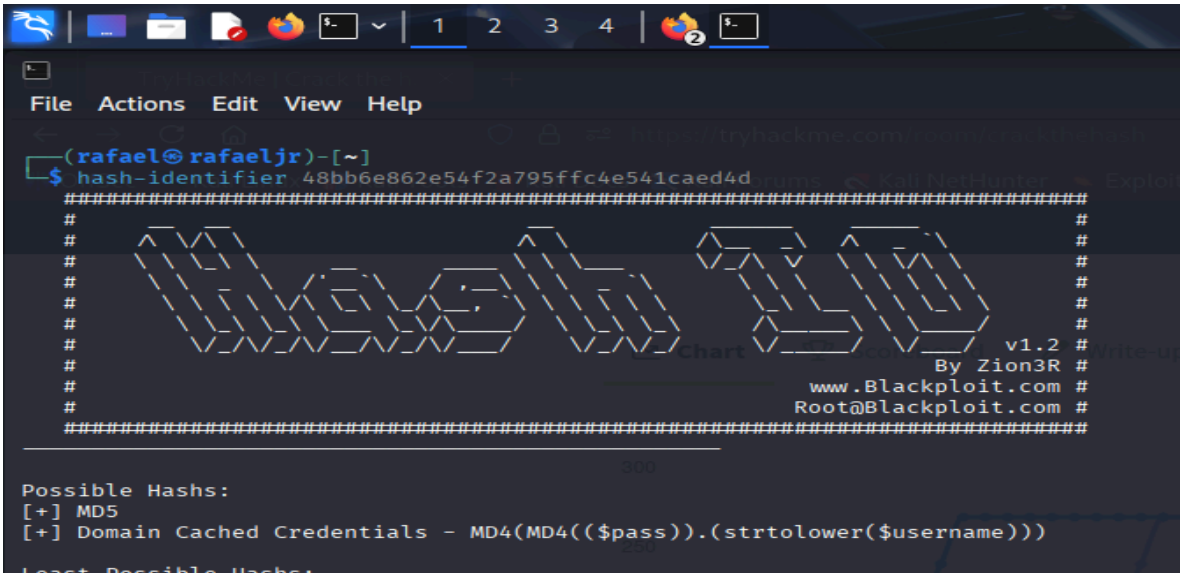
Writeup - CrackTheHash

Rafael Rodrigues Ribeiro Junior

Level 1:
Começando temos a primeira hash: 48bb6e862e54f2a795ffc4e541caed4d

Utilizei o hash-identifier para identificar qual criptografia foi utilizada, ele me deu o resultado MD5



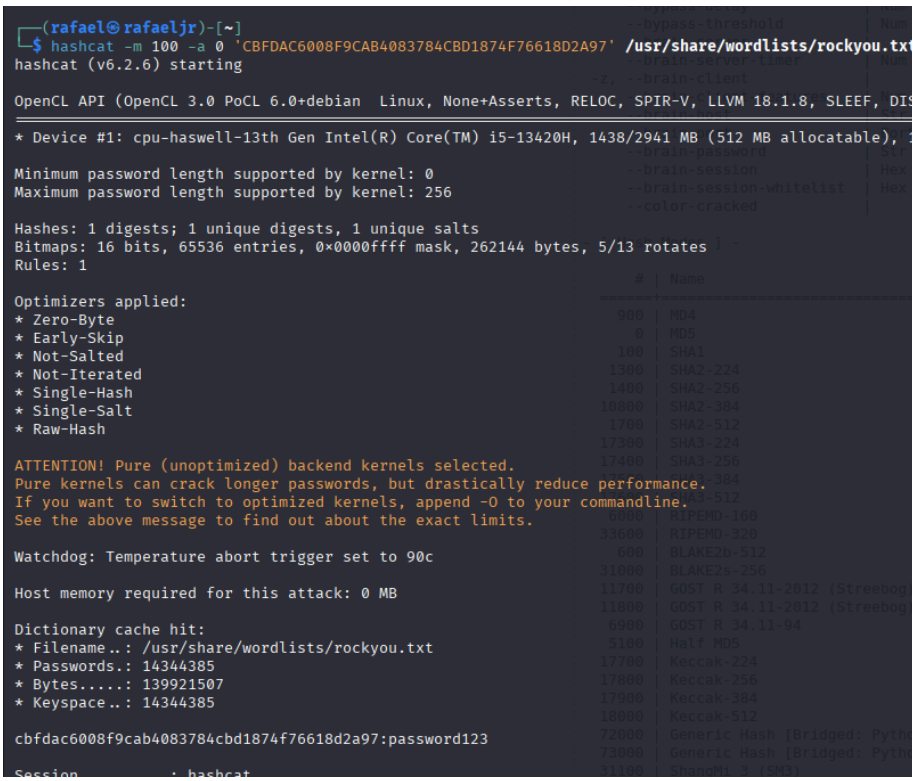
Depois utilizei o hashcat para quebrar a criptografia da hash junto com a wordlist da rockyou, consegui a resposta: easy

```
Watchdog: Temperature abort trigger set to 90c

Host memory required for this attack: 0 MB

Dictionary cache built:
* Filename..: /usr/share/wordlists/rockyou.txt
* Passwords.: 14344392
* Bytes.....: 139921507
* Keyspace..: 14344385
* Runtime...: 2 secs

48bb6e862e54f2a795ffc4e541caed4d:easy

Session..........: hashcat
Status...........: Cracked
Hash.Mode........: 0 (MD5)
Hash.Target......: 48bb6e862e54f2a795ffc4e541caed4d
Time.Started.....: Fri Apr 10 09:21:31 2026 (0 secs)
Time.Estimated...: Fri Apr 10 09:21:31 2026 (0 secs)
Kernel.Feature...: Pure Kernel
Guess.Base.......: File (/usr/share/wordlists/rockyou.txt)
Guess.Queue......: 1/1 (100.00%)
Speed.#1.........:  3137.8 kH/s (0.03ms) @ Accel:256 Loops:1 Thr:1 Vec:8
Recovered........: 1/1 (100.00%) Digests (total), 1/1 (100.00%) Digests (new)
Progress.........: 172544/14344385 (1.20%)
Rejected.........: 0/172544 (0.00%)
Restore.Point....: 172288/14344385 (1.20%)
Restore.Sub.#1...: Salt:0 Amplifier:0-1 Iteration:0-1
Candidate.Engine.: Device Generator
Candidates.#1....: emmabear → dodos
Hardware.Mon.#1..: Util: 98%

Started: Fri Apr 10 09:21:27 2026
Stopped: Fri Apr 10 09:21:32 2026

┌──(rafael㉿rafaeljr)-[~]
```

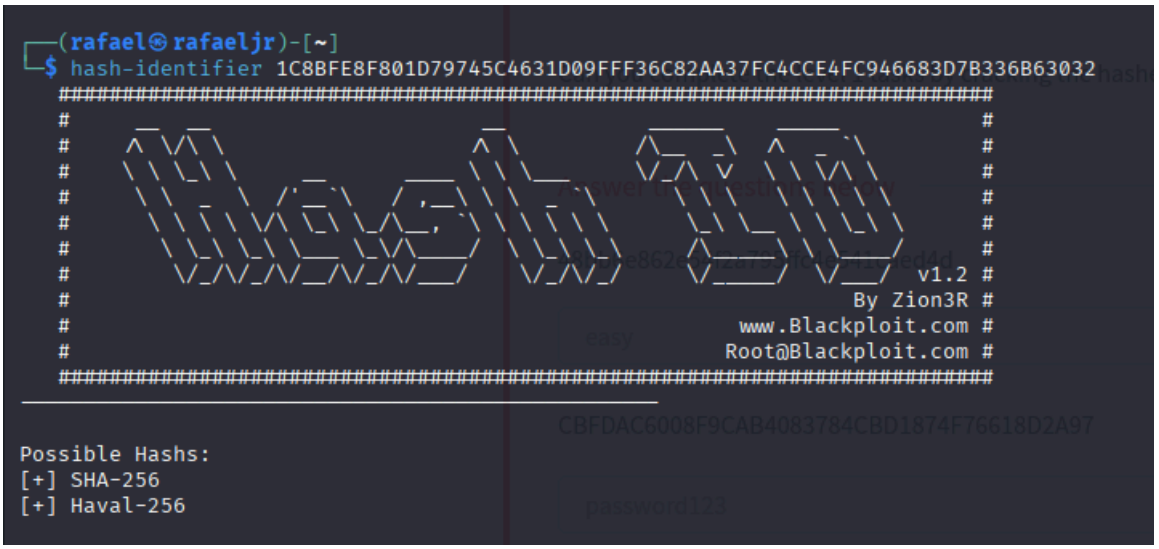
Segunda hash: CBFDAC6008F9CAB4083784CBD1874F76618D2A97
Criptografia: SHA-1

```
┌──(rafael㉿rafaeljr)-[~]
└─$ hash-identifier CBFDAC6008F9CAB4083784CBD1874F76618D2A97
   ##########################################################################
   #     __  __                     __          _____   _____          #
   #    /\ \/\ \                  /\ \        /\__  _\  /\  _ `\        #
   #    \ \ \_\ \     __      ____ \ \ \___    \/_/\ \/  \ \ \/\ \       #
   #     \ \  _  \  /'__`\   / ,__\ \ \  _ `\     \ \ \   \ \ \ \ \      #
   #      \ \ \ \ \/\ \_\ \_/\__, `\ \ \ \ \ \     \_\ \__ \ \ \_\ \      #
   #       \ \_\ \_\ \___ \_\/\____/  \ \_\ \_\    /\_____\ \ \____/      #
   #        \/_/\/_/\/__/\/_/\/___/    \/_/\/_/    \/_____/  \/___/  v1.2 #
   #                                                          By Zion3R #
   #                                                  www.Blackploit.com #
   #                                                 Root@Blackploit.com #
   ##########################################################################
_____________________________________________________

Possible Hashs:
[+] SHA-1
[+] MySQL5 - SHA-1(SHA-1($pass))
```

Resultado: password123



Terceira hash:
1C8BFE8F801D79745C4631D09FFF36C82AA37FC4CCE4FC946683D7B336B63032
Criptografia: SHA-256



Resultado: letmein

(Como eu já tinha feito essa quebra antes, o resultado ficou guardado)

```
┌──(rafael㉿rafaeljr)-[~]
└─$ hashcat -m 1400 -a 0 '1C8BFE8F801D79745C4631D09FFF36C82AA37FC4CCE4FC946683D7B336B63032' /usr/share/wordlists/rockyou.txt --show
1c8bfe8f801d79745c4631d09fff36c82aa37fc4cce4fc946683d7b336b63032:letmein
```

Quarta hash: $2y$12$Dwt1BZj6pcyc3Dy1FWZ5ieeUznr71EeNkJkUlypTsgbX1H68wsRom

Criptografia: ´Não consegui descobrir essa criptografia com o hash-identifier, mas olhando na documentação do hashcat, consegui achar uma criptografia parecida: Bcrypt

3200	bcrypt $2*$, Blowfish (Unix)	$2a$05$LhayLxezLhK1LhWvKxCyLOj0j1u.Kj0jZ0pEmm134uzrQlFvQJLF6

Resultado: bleh

Na primeira abordagem utilizei um bruteforce sem wordlist apenas filtrando por 4 letras minúsculas:

```
┌──(rafael㉿rafaeljr)-[~]
└─$ hashcat -m 3200 -a 3 '$2y$12$Dwt1BZj6pcyc3Dy1FWZ5ieeUznr71EeNkJkUlypTsgbX1H68wsRom' ?l?l?l?l
hashcat (v6.2.6) starting

OpenCL API (OpenCL 3.0 PoCL 6.0+debian  Linux, None+Asserts, RELOC, SPIR-V, LLVM 18.1.8, SLEEF, DISTRO, POCL_DEBUG) - Platform #1 [The pocl project]
* Device #1: cpu-haswell-13th Gen Intel(R) Core(TM) i5-13420H, 1438/2941 MB (512 MB allocatable), 1MCU
```

Mas como estava demorando demais, utilizei a abordagem de filtrar apenas as palavras da rockyou que eram formada por 4 letras minúsculas:

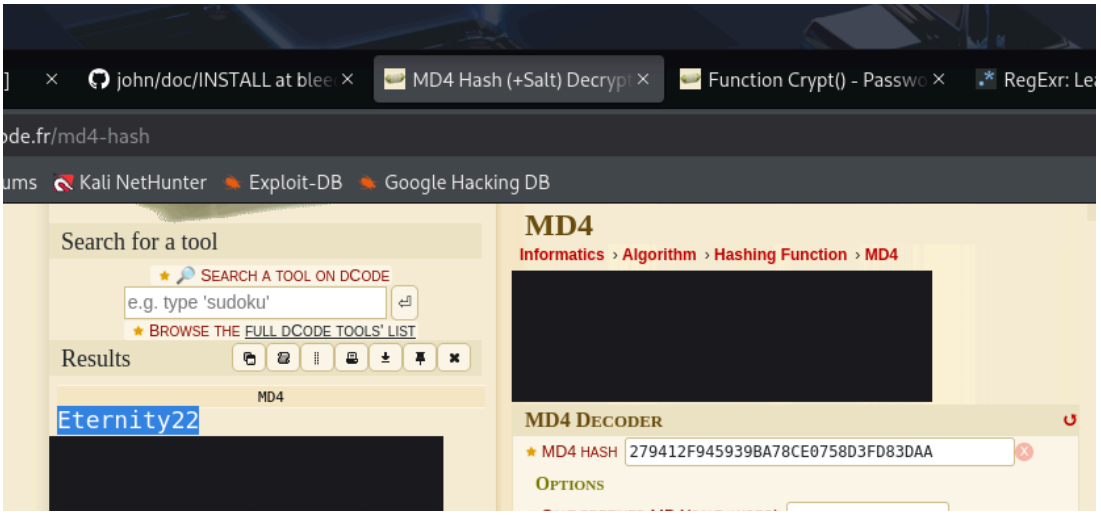
```
┌──(rafael㉿rafaeljr)-[~]
└─$ grep -E '^[a-z]{4}$' /usr/share/wordlists/rockyou.txt > list2.txt
```

```
┌──(rafael㉿rafaeljr)-[~]
└─$ hashcat -m 3200 -a 0 '$2y$12$Dwt1BZj6pcyc3Dy1FWZ5ieeUznr71EeNkJkUlypTsgbX1H68wsRom' list2.txt --show
$2y$12$Dwt1BZj6pcyc3Dy1FWZ5ieeUznr71EeNkJkUlypTsgbX1H68wsRom:bleh
```

Quinta e última hash do level 1: 279412f945939ba78ce0758d3fd83daa
Criptografia: MD4

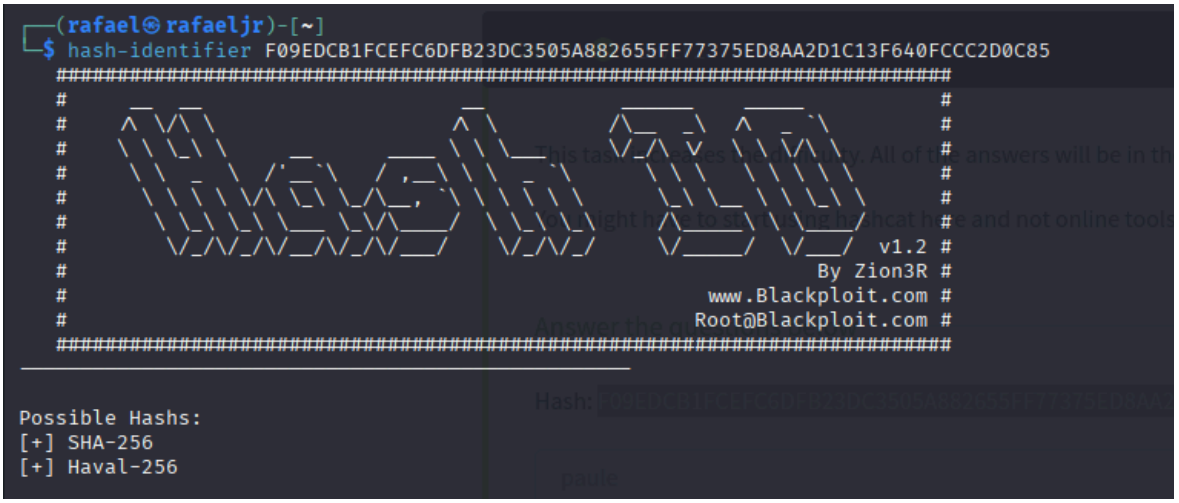
Resultado: Eternity22

Como essa hash era mais fraca resolvi ela utilizando o site dcode:



Level 2:
Primeira hash:
F09EDCB1FCEFC6DFB23DC3505A882655FF77375ED8AA2D1C13F640FCCC2D0C85

Criptografia: SHA-256



Resultado: paule

Segunda hash: 1DFECA0C002AE40B8619ECF94819CC1B
Criptografia: NTLM

O hash-identifier não deu como possível criptografia a NTLM, mas olhando a dica no tryhackme, descobri a criptografia

```
┌──(rafael㉿rafaeljr)-[~]
└─$ hashcat -m 1400 -a 0 'F09EDCB1FCEFC6DFB23DC3505A882655FF77375ED8AA2D1C13F640FCCC2D0C85' /usr/share/wordlists/rockyou.txt --show
f09edcb1fcefc6dfb23dc3505a882655ff77375ed8aa2d1c13f640fccc2d0c85:paule

┌──(rafael㉿rafaeljr)-[~]
└─$ hash-identifier 1DFECA0C002AE40B8619ECF94819CC1B
   ##########################################################################
   #     __  __                  __          _____   ____                 #
   #    /\ \/\ \                /\ \        /\__  _\  /\  _`\             #
   #    \ \ \_\ \     ____ \ \ \___     \/_/\ \/  \ \ \/\ \            #
   #     \ \  _  \  /'__`\   / ,__\ \ \  _ `\      \ \ \   \ \ \ \ \          #
   #      \ \ \ \ \/\ \_\ \_/\__, `\ \ \ \ \ \      \_\ \__ \ \ \_\ \          #
   #       \ \_\ \_\ \___ \_\/\____/  \ \_\ \_\     /\_____\ \ \____/          #
   #        \/_/\/_/\/__/\/_/\/___/    \/_/\/_/     \/_____/  \/___/  v1.2 #
   #                                                          By Zion3R #
   #                                                  www.Blackploit.com #
   #                                                 Root@Blackploit.com #
   ##########################################################################
_____________________________________________________

Possible Hashs:
[+] MD5
[+] Domain Cached Credentials - MD4(MD4(($pass)).(strtolower($username)))

Least Possible Hashs:
[+] RAdmin v2.x
[+] NTLM
```

Resultado:

```
┌──(rafael㉿rafaeljr)-[~]
└─$ hashcat -m 1000 -a 0 '1DFECA0C002AE40B8619ECF94819CC1B' /usr/share/wordlists/rockyou.txt --show
1dfeca0c002ae40b8619ecf94819cc1b:n63umy8lkf4i
```

Terceira hash:
6aReallyHardSalt$6WKUTqzq.UQQmrm0p/T7MPpMbGNnzXPMAXi4bJMl9be.cfi3/qxIf.hs
Criptografia: Sha512Crypt

Também olhando a documentação do hashcat encontrei uma hash parecida:

1800	sha512crypt 6, SHA512 (Unix) [2]	6qdMgClgO2dQWB37F$jhexCX1SdsCAi0OZmoRVAPnWSwuP/mHVhXIMJfKlaacxFkwWLDZ0ViF8Ur3WcHashcatVp2WShcElLi8QZCbt/
2000	STDOUT	n/a

Resultado: waka99

Também resolvi essa filtrando as palavras da rockyou com 6 letras

```
┌──(rafael㉿rafaeljr)-[~]
└─$ hashcat -m 1800 -a 0 '$6$aReallyHardSalt$6WKUTqzq.UQQmrm0p/T7MPpMbGNnzXPMAXi4bJMl9be.cfi3/qxIf.hsGpS41BqMhSrHVXgMpdjS6xeKZAs02.' list5.txt --show
$6$aReallyHardSalt$6WKUTqzq.UQQmrm0p/T7MPpMbGNnzXPMAXi4bJMl9be.cfi3/qxIf.hsGpS41BqMhSrHVXgMpdjS6xeKZAs02.:waka99
```

Quarta hash: e5d8870e5bdd26602cab8dbe07a942c8669e56d6
Salt: tryhackme
Criptografia: HMAC-SHA1

```
┌──(rafael㉿rafaeljr)-[~]
└─$ hash-identifier e5d8870e5bdd26602cab8dbe07a942c8669e56d6
   #########################################################################
   #     __  __                     __      _____   ___  ___  ___ ___  ___ #
   #    /__\/__\ /__\   /\/\/\/\/\/\/\/\/\/\/\/\/\/\/\/\/\/\/\/\/\/\/\/\/\ #
   #    \_\ \_\ \_\ \   \ \ \ \ \ \ \ \ \ \ \ \ \ \ \ \ \ \ \ \ \ \ \ \ \ #
   #     \ \ \_/ \ \ \   \ \ \ \ \ \ \ \ \ \ \ \ \ \ \ \ \ \ \ \ \ \ \ \ \ #
   #      \ \ \ \ \ \ \   \ \ \ \ \ \ \ \ \ \ \ \ \ \ \ \ \ \ \ \ \ \ \ \ \ #
   #       \ \ \ \ \ \ \   \ \ \ \ \ \ \ \ \ \ \ \ \ \ \ \ \ \ \ \ \ \ \ \ \ #
   #                                                                       #
   #                                                             By Zion3R #
   #                                                    www.Blackploit.com #
   #                                                   Root@Blackploit.com #
   #########################################################################
--------------------------------------------------
Possible Hashs:
[+] SHA-1
[+] MySQL5 - SHA-1(SHA-1($pass))
```

O hash-identifier identificou a hash como SHA-1, mas como ela tinha Salt procurei uma combinação e encontrei o HMAC-SHA1, rodando no hashcat, descobri a senha

Resultado:

```
File  Actions  Edit  View  Help

┌──(rafael㉿rafaeljr)-[~]
└─$ hashcat -m 160 -a 0 e5d8870e5bdd26602cab8dbe07a942c8669e56d6:tryhackme /usr/share/wordlists/rockyou.txt
hashcat (v6.2.6) starting

OpenCL API (OpenCL 3.0 PoCL 6.0+debian  Linux, None+Asserts, RELOC, SPIR-V, LLVM 18.1.8, SLEEF, DISTRO, POCL_DEB

* Device #1: cpu-haswell-13th Gen Intel(R) Core(TM) i5-13420H, 1438/2941 MB (512 MB allocatable), 1MCU

Minimum password length supported by kernel: 0
Maximum password length supported by kernel: 256

Hashes: 1 digests; 1 unique digests, 1 unique salts
Bitmaps: 16 bits, 65536 entries, 0x0000ffff mask, 262144 bytes, 5/13 rotates
Rules: 1

Optimizers applied:
* Zero-Byte
* Not-Iterated
* Single-Hash
* Single-Salt

ATTENTION! Pure (unoptimized) backend kernels selected.
Pure kernels can crack longer passwords, but drastically reduce performance.
If you want to switch to optimized kernels, append -O to your commandline.
See the above message to find out about the exact limits.

Watchdog: Temperature abort trigger set to 90c

Host memory required for this attack: 0 MB

Dictionary cache hit:
* Filename..: /usr/share/wordlists/rockyou.txt
* Passwords.: 14344385
* Bytes.....: 139921507
* Keyspace..: 14344385

[s]tatus [p]ause [b]ypass [c]heckpoint [f]inish [q]uit => s

Session..........: hashcat
Status...........: Running
Hash.Mode........: 160 (HMAC-SHA1 (key = $salt))
Hash.Target......: e5d8870e5bdd26602cab8dbe07a942c8669e56d6:tryhackme
Time.Started.....: Wed Apr 22 19:32:22 2026 (5 secs)
Time.Estimated...: Wed Apr 22 19:32:28 2026 (1 sec)
Kernel.Feature...: Pure Kernel
Guess.Base.......: File (/usr/share/wordlists/rockyou.txt)
Guess.Queue......: 1/1 (100.00%)
Speed.#1.........:  2141.8 kH/s (0.09ms) @ Accel:256 Loops:1 Thr:1 Vec:8
Recovered........: 0/1 (0.00%) Digests (total), 0/1 (0.00%) Digests (new)
Progress.........: 10237184/14344385 (71.37%)
Rejected.........: 0/10237184 (0.00%)
Restore.Point....: 10237184/14344385 (71.37%)
Restore.Sub.#1...: Salt:0 Amplifier:0-1 Iteration:0-1
Candidate.Engine.: Device Generator
Candidates.#1....: alfi91 -> alfathy
Hardware.Mon.#1..: Util:100%

e5d8870e5bdd26602cab8dbe07a942c8669e56d6:tryhackme:481616481616
```